

EE3408C

- The following topics will be covered in this module:
 - MOSFET characteristic and small-signal model
 - Single/multi-stage amplifier design
 - Frequency response and compensation
 - Current sources and mirrors
 - Differential amplifier analysis & design
 - Operational amplifier analysis & design
 - Bias circuits

Week	Tutorial (Tuesday 12pm – 1pm) LT4	Lecture (Thursday 12pm – 2pm) LT3	Remark
WK1 (11-15 Aug)	- Module Introduction	- Review of Basic Semiconductor Physics and MOSFET Characteristics	
WK2 (18-22 Aug)	Open Lab	- MOSFET Small-Signal Model	
WK3 (25-29 Aug)	Workshop1 on EDA	- Single-Stage Amplifier Part 1	Project Starts
WK4 (1-5 Sep)	Workshop2 on EDA	- Single-Stage Amplifier Part 2	
WK5 (8-12 Sep)	Q&A for EDA	- Frequency Response (1)	
WK6 (15-19 Sep)	Tutorial-1	- Current Sources & Mirrors	Sub-project 1 report due

Recess Week 20 Sep 2025 – 28 Sep 2025

- EDA workshop is not at LT4

Week	Tutorial (Tuesday 12pm – 1pm) LT4	Lecture (Thursday 12pm – 2pm) LT3	Remark
WK7 (29 Sep - 3 Oct)	Q&A	- Differential Pair Part 1	
WK8 (6-10 Oct)	Tutorial-2	- Differential Pair Part 2 - Operational Amplifier Part 1	
WK9 (13-17 Oct)	Mid-term quiz	- Operational Amplifier Part 2	
WK10 (20 – 24 Oct)	Tutorial-3	- Frequency Response (2) Part 1	Sub-project 2 report due
WK11 (27-31 Oct)	Q&A	- Frequency Response (2) Part 2	
WK12 (3-7 Nov)	Tutorial-4	- Bias Circuit	
WK13 (10-14 Nov)	Backup and review	Backup and review	Sub-project 3 report due

- Assignment ----- 20%
- Mid-term Quiz ----- 10%
- Design project ----- 30%
 - Work in a team of two students
 - Declare contribution of each team member in the report
- Final exam ----- 40%

(1 A4 sized paper with **handwritten notes** on both sides is allowed for quiz and final exam)

- Behzad Razavi, *Design of Analog CMOS Integrated Circuits*, McGRAW-HILL, 1st or 2nd International edition, 2000/2017
- Allen & Holberg, *CMOS Analog Circuit Design*, 2nd or 3rd Ed. Oxford University Press, 2011
- R J Baker, H. W. Li and D. E. Boyce, “*CMOS circuit design, layout and simulation*”, 3rd or 4th Edition, IEEE press, 2008/2019

